

```
In [1]: import pandas as pd

df = pd.read_csv("usarrests.csv")

print("--- Data Preview ---")
display(df.head())

print("\n--- Summary Statistics ---")
display(df.describe())
```

```
--- Data Preview ---
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody
tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th>
<th>Unnamed: 0</th> <th>Murder</th> <th>Assault</th> <th>UrbanPop</th> </thead>
<tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody>
</tbody>
```

Alabama	13.2	236.0	58
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Alaska	10.0	263.0	48
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Arizona	8.1	294.0	80
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Arkansas	8.8	190.0	50
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California	9.0	276.0	91
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```
--- Summary Statistics ---
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<th>Murder</th> <th>Assault</th> <th>UrbanPop</th> </thead> <tbody>
<th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%
</th> <th>75%</th> <th>max</th> </tbody> <tbody> </tbody>
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50.00000	49.000000	50.00000
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7.78800	182.183673	74.20000
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4.35551	130.877435	73.40828
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0.80000	45.000000	6.00000
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4.07500	109.000000	53.25000
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7.25000	159.000000	66.00000
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11.25000	249.000000	77.75000
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17.40000	879.000000	570.00000
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```
In [2]: print("--- Checking for Impossible Urban Population (> 100) ---")
impossible_urban = df[df["UrbanPop"] > 100]
display(impossible_urban)
```

```
print("\n--- Checking for Extreme Crime Rates ---")
extreme_crime = df[(df["Assault"] > 300) | (df["Murder"] > 20)]
display(extreme_crime)
```

--- Checking for Impossible Urban Population (> 100) ---

```
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<tbody> <th>14</th> </tbody> <tbody></tbody>
```

```
Iowa 2.2 56.0 570
```

--- Checking for Extreme Crime Rates ---

```
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<th>Unnamed: 0</th> <th>Murder</th> <th>Assault</th> <th>UrbanPop</th> </thead>
<tbody> <th>8</th> <th>32</th> <th>39</th> </tbody> <tbody></tbody>
```

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Florida 15.4 335.0 80
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```
North Carolina 13.0 337.0 45
```

```
South Carolina 14.4 879.0 48
```

<p>Outlier Investigation Log: US Arrests Data Analysis: Impossible vs. Extreme</p> <p>For Impossible Values (UrbanPop > 100%): Given that we're looking at a population of people, the maximum value we can have is 100%, so any value over 100 is impossible as opposed to an outlier, and is more likely to be an error or evidence of corruption. As a result, any rows showing a value over 100 should be investigated and corrected if possible. They should be dropped if they cannot be corrected, as known mistakes should not influence other data.</p> <p>For Extreme Values (Extremely High Assault/Murder Rates): In contrast, extremely high arrest rates (per 100,000 residents), while rare, are not impossible. Extreme values should not be automatically deleted, as that can destroy legitimate variance and ruin bias checks. As a result, these rows should be retained, but carefully documented to show that they represent outliers, not mistakes.</p>